

Experiment-24

Aim

To perform image sharpening using a High-Boost Mask and enhance the edges and fine details of the input image.

Procedure

1. Select and load the input image.
2. Convert the image to grayscale.
3. Choose a high-boost factor A , for example $A=2$.
4. Create the high-boost mask.
5. Apply the mask to the image using convolution.
6. Clip the resulting pixel values to the range 0–255.
7. Display the original and sharpened images.
8. Compare the images and observe the enhancement of edges and fine details.

Python Code

```
import cv2
import numpy as np
img = cv2.imread("image.jpg")
if img is None:
    print("Error: Image not found")
    exit()
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
A = 2
mask = np.array([
    [0, -1, 0],
    [-1, A + 4, -1],
    [0, -1, 0]
], dtype=np.float32)
sharpened = cv2.filter2D(gray, -1, mask)
sharpened = np.clip(sharpened, 0, 255).astype(np.uint8)
```

```
cv2.imshow("Original Image", gray)
cv2.imshow("High-Boost Sharpened Image", sharpened)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Sample Input:



Output:



Results:

The input image was successfully sharpened using a High-Boost Mask. The high-boost operation enhanced the edges, boundaries, and fine details of the image. Increasing the value of A produces a stronger sharpening effect.